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UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

CANDIDATE NAME

CENTRE NUMBER | | | | | |

CANDIDATE NUMBER | | | | |

4848904618

ADDITIONAL MATHEMATICS **4037/22**

Paper 2 **May/June 2011**

2 hours

Candidates answer on the Question Paper.

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use a pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Answer **all** the questions.
Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.
The use of an electronic calculator is expected, where appropriate.
You are reminded of the need for clear presentation in your answers.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.
The total number of marks for this paper is 80.

For Examiner's Use	
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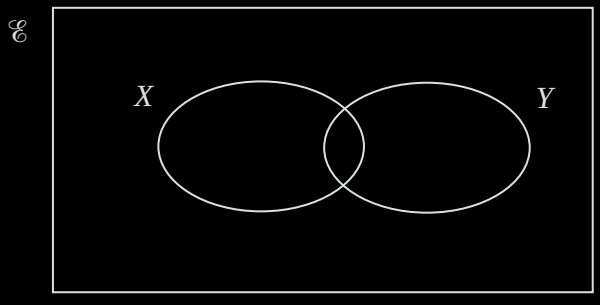
This document consists of **16** printed pages.

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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

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On the diagram shade the region which represents the set $X \cup Y'$. [1]

(ii) Using set notation express the set $X \cup Y'$ in an alternative way. [1]

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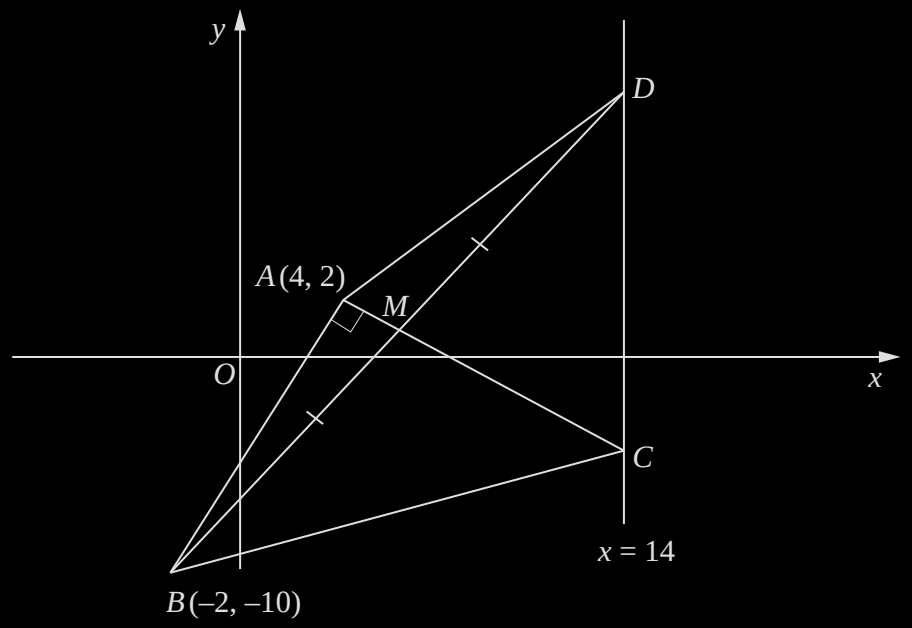
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The diagram shows the quadrilateral *ABCD* in which *A* is the point (4, 2) and *B* is the point (–2, –10). The points *C* and *D* lie on the line $x = 14$. The diagonal *AC* is perpendicular to *AB* and passes through the mid-point, *M*, of the diagonal *BD*. Find the area of the quadrilateral *ABCD*.

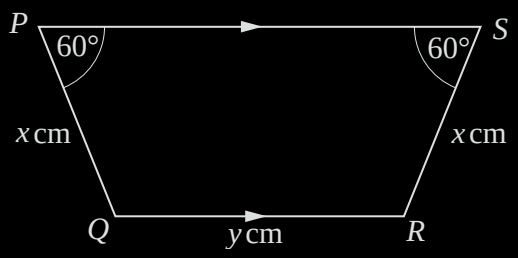
[9]

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PQRS is a trapezium in which $PQ = RS = x$ cm and $QR = y$ cm.
Angle QPS = angle $RSP = 60°$ and QR is parallel to PS.

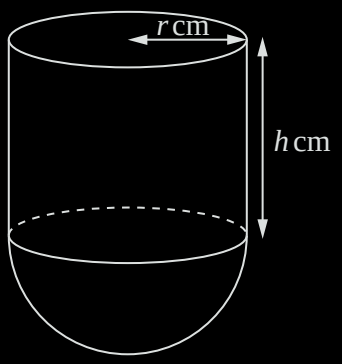
(i) Given that the perimeter of the trapezium is 60 cm, express y in terms of x. [2]

(ii) Given that the area of the trapezium is A cm^2, show that

$$A = \frac{\sqrt{3}(30x - x^2)}{2}.$$ [3]

(iii) Given that x can vary, find the value of x for which A has a stationary value and determine the nature of this stationary value. [4]

OR



For a sphere of radius r:

Volume $= \frac{4}{3}\pi r^3$

Surface area $= 4\pi r^2$

The diagram shows a solid object in the form of a cylinder of height h cm and radius r cm on top of a hemisphere of radius r cm. Given that the volume of the object is 2880π cm^3,

(i) express h in terms of r, [2]

(ii) show that the external surface area, A cm^2, of the object is given by

$$A = \frac{5}{3}\pi r^2 + \frac{5760\pi}{r}.$$ [3]

Given that r can vary,

(iii) find the value of r for which A has a stationary value, [4]

(iv) find this stationary value of A, leaving your answer in terms of π, [2]

(v) determine the nature of this stationary value. [1]

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